Official Hugging Face notebook: vit-base-patch16-224-in21k
Notebook run in Google Colab, with its saved outputs. Opened from: https://huggingface.co/google/vit-base-patch16-224-in21k.ipynb

In [1]:
```
!pip install -U transformers
```

```
Requirement already satisfied: transformers in
/usr/local/lib/python3.13/dist-packages (5.20.2)
Requirement already satisfied: huggingface-hub<3.0,>=1.31.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (1.33.0)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.13/dist-packages (from transformers) (2.1.3)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (26.3)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in
/usr/local/lib/python3.13/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.23.3)
Requirement already satisfied: typer in /usr/local/lib/python3.13/dist-
packages (from transformers) (0.27.3)
Requirement already satisfied: safetensors>=0.8.0 in
/usr/local/lib/python3.13/dist-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in
/usr/local/lib/python3.13/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (8.5.0)
Requirement already satisfied: filelock>=3.10.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (4.2.4)
Requirement already satisfied: fsspec>=2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (0.28.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-
hub<3.0,>=1.31.0->transformers) (4.16.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (4.15.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (2026.10.19)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<3.0,>=1.31.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-
packages (from httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0-
>transformers) (3.20)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.13/dist-packages (from httpcore==1.*-
```

```
>httpx<1,>=0.23.0->huggingface-hub<3.0,>=1.31.0->transformers) (0.16.0)
Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers)
(13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.13/dist-packages (from typer->transformers) (0.0.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer-
>transformers) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0-
>rich>=13.8.0->typer->transformers) (0.1.2)
```

Local Inference on GPU

Model page: https://huggingface.co/google/vit-base-patch16-224-in21k

⚠️ If the generated code snippets do not work, please open an issue on either the model repo and/or on huggingface.js 🙏

In [2]:
```python
# Use a pipeline as a high-level helper
from transformers import pipeline

pipe = pipeline("image-feature-extraction", model="google/vit-base-
patch16-224-in21k")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please
set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable
higher rate limits and faster downloads.
```

config.json: 100% 502/502 [00:00<00:00, 18.9kB/s]

model.safetensors: reconstructing file: 100% 346MB / 346MB, 27.2MB/s

model.safetensors: downloading bytes: 328MB, 23.7MB/s

Loading weights: 100% 200/200 [00:00<00:00, 3394.47it/s]

preprocessor_config.json: 100% 160/160 [00:00<00:00, 3.19kB/s]

In [3]:
```python
# pip install -U transformers accelerate
# Load model directly
from transformers import AutoImageProcessor, AutoModel

processor = AutoImageProcessor.from_pretrained("google/vit-base-patch16-
224-in21k")
model = AutoModel.from_pretrained("google/vit-base-patch16-224-in21k",
device_map="auto")
```

Loading weights: 100% 200/200 [00:00<00:00, 3151.41it/s]